LAB PRACTICAL # 02

Basic Linux Commands

Name	Manahil Zahra
Roll Number	F24-2310
Section	D
Program	BS-SE (5th Semester)

1. Objective

The objectives of this lab are:

- To learn the basic syntax of Linux commands.
- To display the current date and time.
- To display the calendar of a specific year and month.
- To check the available disk storage and memory.
- To display information about users.
- To learn how to access the manual pages of Linux commands.

2. Software Required

- Linux Operating System
- Linux Terminal

3. Lab Activity 1: Basic Linux Commands

The following tasks were performed in the Linux terminal.

Task	Description	Command	Purpose
1	Display the current date	`date`	Displays the current system date and time.
2	Display the calendar for the current year	`cal 2026`	Displays the complete calendar for the year 2026, including all twelve months.
3	Display the calendar of 2012	`cal 2012`	Displays the complete calendar for the year 2012.
4	Display the calendar of February 2015	`cal 2 2015`	Displays the calendar for February 2015. The number 2 represents February and 2015 is the year.

Task	Description	Command	Purpose
5	Display the free storage on the machine	`df`	Displays disk storage information, including total, used, and available space on the file systems.
6	Display the free memory on the machine	`free`	Displays memory information, including total, used, and available memory.
7	Display the user information	`who`	Displays information about users currently logged in to the computer.
8	Open the manual pages of date and free	`man date` `man free`	Displays the manual page of a command, with its description, options, and usage. Press q to exit the manual page.

4. Important Linux Commands

The table below summarizes the commands used in this lab.

Command	Purpose
`date`	Displays the current date and time
`cal`	Displays a calendar for a year or month
`df`	Displays disk storage information
`free`	Displays memory usage information
`exit`	Ends the current shell session
`who`	Displays currently logged-in users
`who am i`	Displays information about the current terminal session
`clear`	Clears the terminal screen
`man`	Displays the manual of a Linux command

5. Basic Syntax of Linux Commands

A Linux command is an instruction given to the computer through the terminal. The shell reads and executes the command.

Commands may contain options and arguments. Options modify the behavior of a command, while arguments specify the item on which the command operates.

Example:

```
date +%H-%M-%S
```

This command displays the time in hours, minutes, and seconds.

6. Result

The basic Linux commands were studied and executed to display the current date, view yearly and monthly calendars, check disk storage and memory, display user information, and access command manuals.

7. Conclusion

In this lab, I learned how to use basic Linux commands in the terminal. I practiced displaying date and calendar information, checking disk space and memory, viewing logged-in users, and using manual pages to understand Linux commands.